

Program 8: Create a Kubernetes pods using YAML dependencies

File structure:

Nodejs-configmap.yaml

Nodejs-pod.yaml

server.js

server.js

```
const http = require("http")
```

```
const port = 3000
```

```
const server = http.createServer((req,res)=>{
```

```
  res.end("Hello from Node.js running")
```

```
});
```

```
server.listen(port,()=>{
```

```
  console.log(`Server is running in port ${port}`);
```

```
})
```

Nodejs-configmap.yaml

```
apiVersion: v1
```

```
kind: ConfigMap
```

```
metadata:
```

```
  name: nodejs-app-config
```

```
data:
```

```
  server.js: |
```

```
    const http = require("http");
```

```
    const port = 3000;
```

```
    const server = http.createServer((req, res) => {
```

```
      res.end("Hello from Node.js running inside kubernetes");
```

```
    });
```

```
    server.listen(port, () => {
```

```
      console.log(`Server is running on port ${port}`);
```

```
    });
```

Nodejs-pod.yaml

```
apiVersion: v1
```

```
kind: Pod
```

```
metadata:
```

```
  name: program-8
```

```
  labels:
```

```
    app: nodejs
```

```
spec:
```

```
  containers:
```

```
    - name: nodejs
```

```
      image: node:18
```

```
command: ["node", "/usr/src/app/server.js"]
volumeMounts:
  - name: app-code
    mountPath: /usr/src/app
ports:
  - containerPort: 3000
volumes:
  - name: app-code
configMap:
  name: nodejs-app-config
```

Start minikube

```
#minikube start
```

Check the status to verify all the services are running

```
#minikube status
```

Create the pod from YAML files

```
#kubectl apply -f nodejs-configmap.yaml
```

```
#kubectl apply -f nodejs-pod.yaml
```

List the pods

```
#kubectl get pods
```

Forward the port to see the nodejs output in browser

```
#kubectl port-forward pod/program-8 3000:3000
```

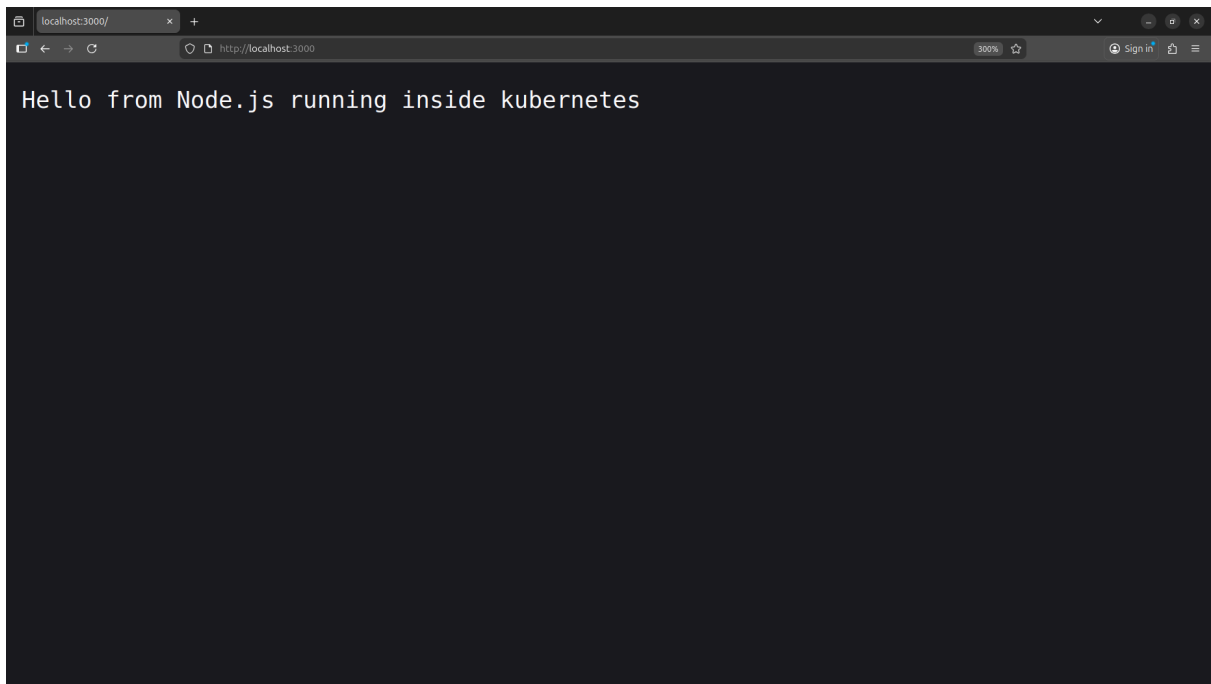


```
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ: ~/Desktop/LabPrograms/Lab8
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/Lab8$ minikube start
🐳 minikube v1.37.0 on Ubuntu 24.04
💡 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
📦 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟 Enabled addons: storage-provisioner, default-storageclass

! /usr/bin/kubectl is version 1.30.14, which may have incompatibilities with Kubernetes 1.34.0.
  ▪ Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
👏 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/Lab8$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/Lab8$ kubectl apply -f nodejs-configmap.yaml
configmap/nodejs-app-config unchanged
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/Lab8$ kubectl apply -f nodejs-pod.yaml
pod/program-8 created
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/Lab8$ kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
program-8     1/1     Running   0           11s
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/Lab8$ kubectl port-forward pod/program-8 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::]:3000 -> 3000
```

Go to browser and search for the url <http://localhost:3000> to see the output



Deleting resources

```
#kubectl delete pods --all
```

```
#kubectl delete svc--all
```

```
#kubectl delete deployments --all
```